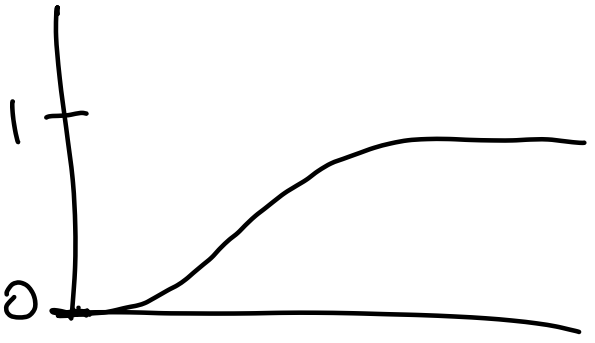
Logistic Regression

- predicts probability of data point belonging to a certain class
- sign = $w \cdot x + b \Rightarrow$ pass through sigmoid in order to get probability
- $f(z) = \frac{1}{1+e^{-z}}$

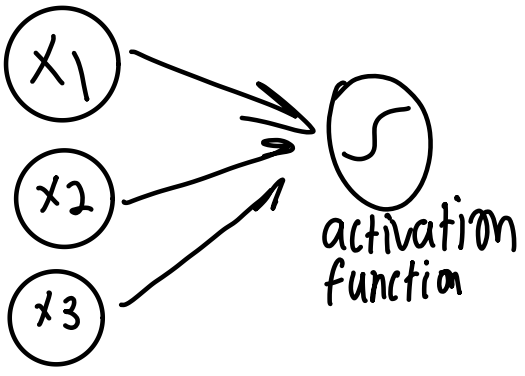


- $z = w \cdot x + b$
- $P(y=1|x) = \frac{1}{1+e^{-(w \cdot x + b)}}$
- $P(y=-1|x) = 1 - \frac{1}{1+e^{-(w \cdot x + b)}}$

$$= \frac{1+e^{-(w \cdot x + b)} - 1}{1+e^{-(w \cdot x + b)}} = \frac{e^{-(w \cdot x + b)}}{1+e^{-(w \cdot x + b)}}$$

Neural Network

- $z = w \cdot x + b$ (perceptron)
- $y = f(z)$ (output of the perceptron)
- activation function is non-linear function that allows nn to learn complex nonlinear patterns in data



Experimental Design

- can vary epochs, cross validation, and learning rate for both models